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MZ 505
Inventory Systems

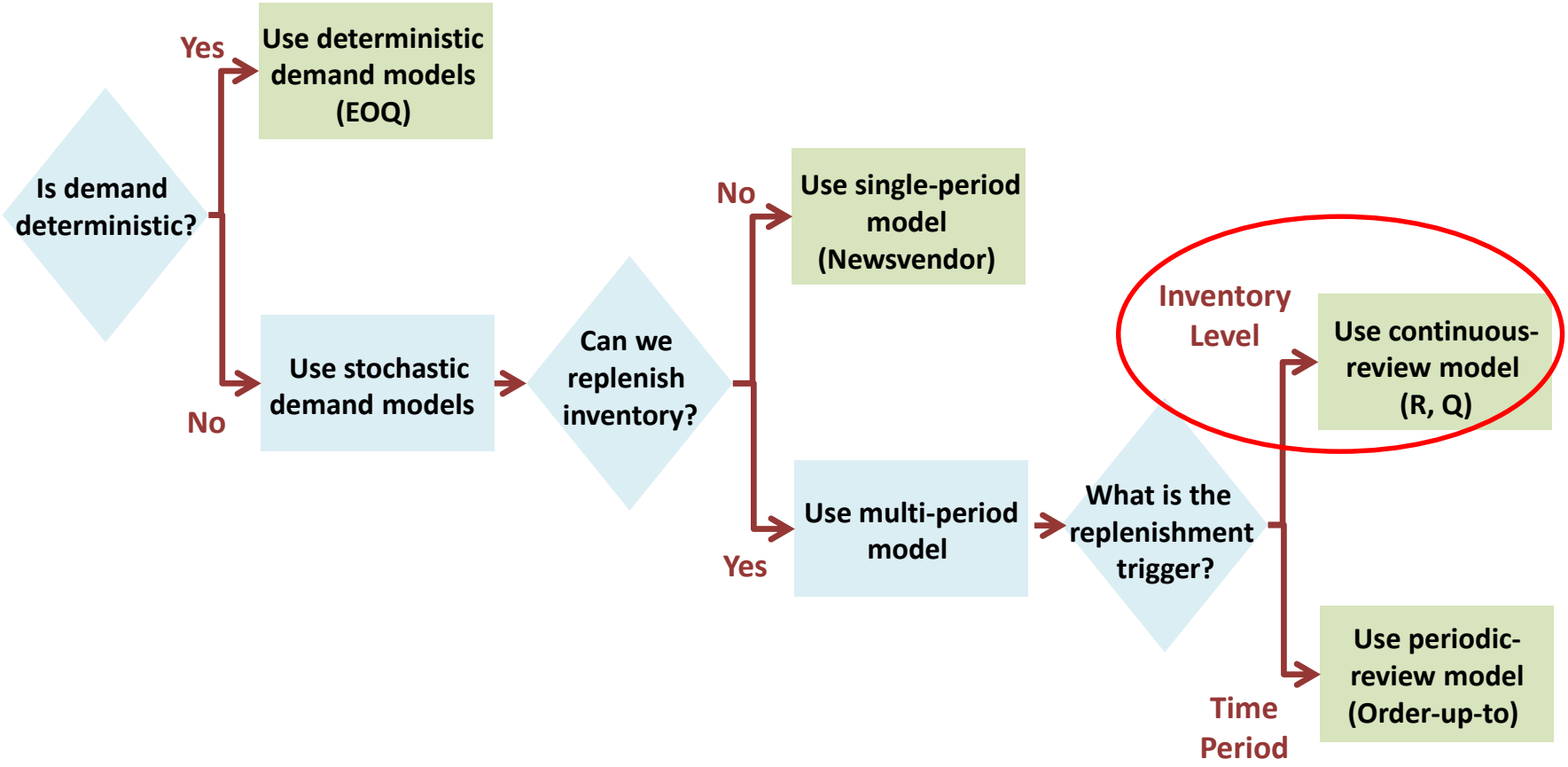
Spyros Lekkakos
slekkakos@zlc.edu.es

Session 17: The Lot Size - Reorder Point (R, Q) Model

Some slides from S. Nahmias, “Production and Operations Analysis“ e09 (2009)

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Decision tree for choosing inventory model



Adapted from Janice H. Hammond, HBP 8016 (2014)

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The (R, Q) Model: Approximate Formulation

(R, Q) Model Assumptions

- Continuous review: Inventory is is being monitored on a continuous basis and replenishment orders can be placed at any time.
- Demand is random and stationary (independent, identically distributed).
  - Expected demand is D per unit of time (i.e., day, week, etc.).
  - Standard deviation is σ
- Lead time, L, is constant
- Unmet demand is backordered
- Cost parameters
  - S: Setup cost per order
  - h: Holding cost per unit of inventory per unit of time
  - b: Backorder cost per unit backordered

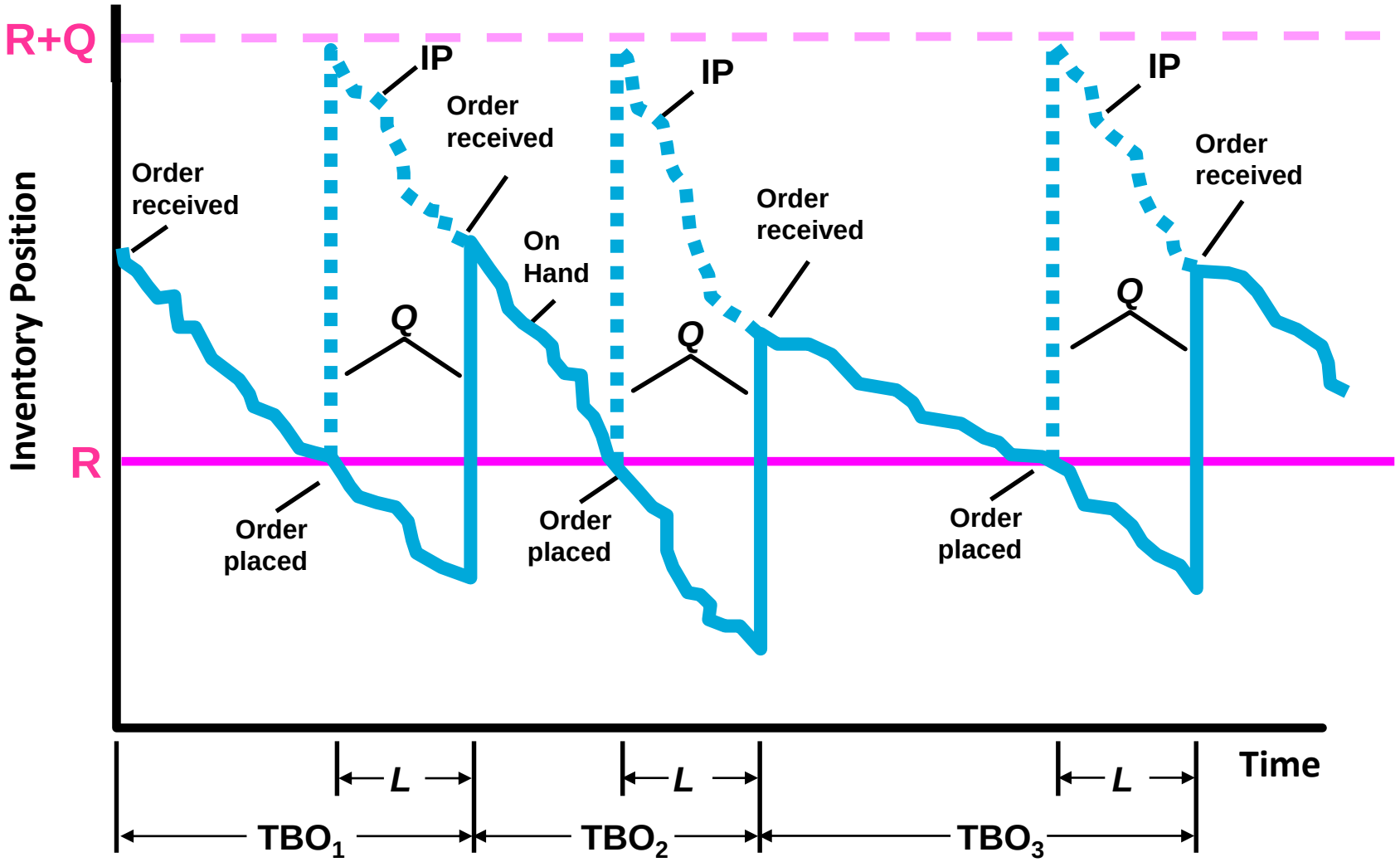
Definitions

- **On-order (pipeline) inventory** = the number of units that have been ordered but have not been received
- **On-hand inventory** = the number of units physically in inventory ready to serve demand
- **Backorder** = the total amount of demand that has not been satisfied:
  - All backordered demand is eventually filled, i.e., there are no lost sales
- **Inventory level (IL)** = On-hand inventory - Backorder
- **Inventory position (IP)** = On-order inventory + Inventory level

(R, Q) Model: Decision variables and policy

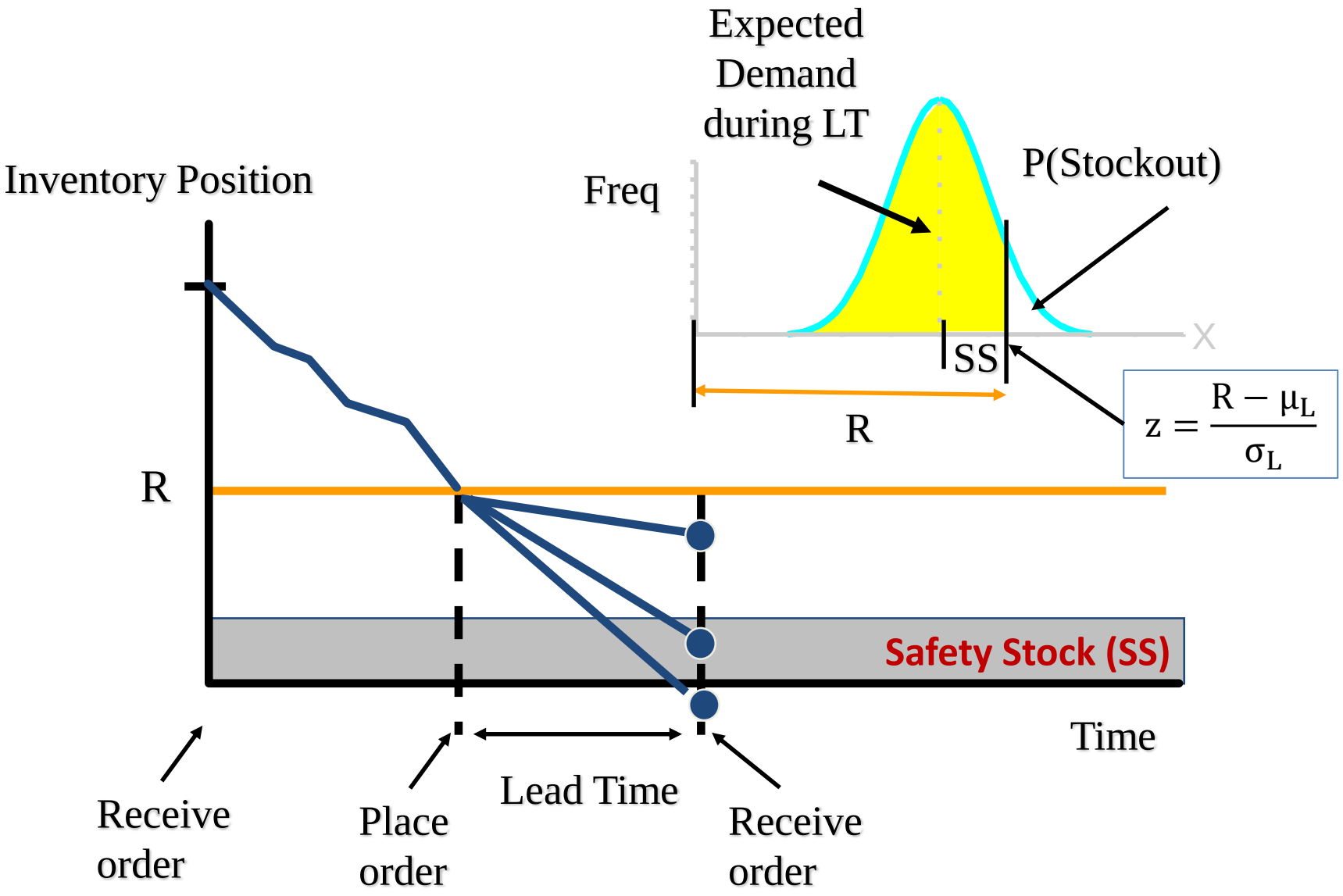
- There are two decision variables in an (R, Q) system:
  - Order quantity, Q and
  - Reorder point, R
- The (R, Q) policy functions as follows:
  - When the inventory position "hits" (or exceeds) the reorder point, R, an order of size Q is placed.
- **Objective**: We want to **choose R and Q** so that expected **total cost** (the sum of expected ordering cost, inventory holding cost, and backorder penalty cost) per unit of time is **minimized**.
- Such policies are often implemented by a point-of-sale computerized system for replenishing a retailer's inventory.

(R, Q) Model: Inventory dynamics



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(R, Q) model: Safety stock & stockout probability



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(R, Q) Model: Summary of notation

L: Replenishment lead time

D: Average demand per unit of time

σ: Standard deviation of demand per unit of time

μ_L: Mean demand during lead time

σ_L: Standard deviation of demand during lead time

CSL: Cycle service level

SS: Safety inventory

R: Reorder point

S: Setup cost per order

h: Holding cost per unit per unit of time

b: Backorder cost per unit backordered

(R, Q) model: Some additional assumptions

- There is never more than a single order outstanding.
  - This assumption is necessary for deriving the approximate cost function that follows.
  - It can be shown that our results also hold if we relax this assumption and allow an arbitrary number of orders to be outstanding.
- The reorder point R (based on the inventory position) is positive.
  - In the simple EOQ model, R is the demand during the lead time.
  - However, in the presence of random demand, R usually includes some safety stock in addition to the expected demand during the lead time.

(R, Q) model: In-stock probability

- Demand during lead time is a random variable, D_L, with
  - PDF(density function) f(x) and CDF (distribution function) F(x)
  - Mean = μ_L and Standard Deviation = σ_L
- Stockout occurs if demand during lead time exceeds the reorder point.
- In-stock probability is the probability that all demand is filled in a cycle:

$$\text{In-stock probability} = \text{Prob}\{D_L \leq R\}$$
$$= 1 - \text{Stockout probability}$$

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(R, Q) model: Safety stock & Expected backorders

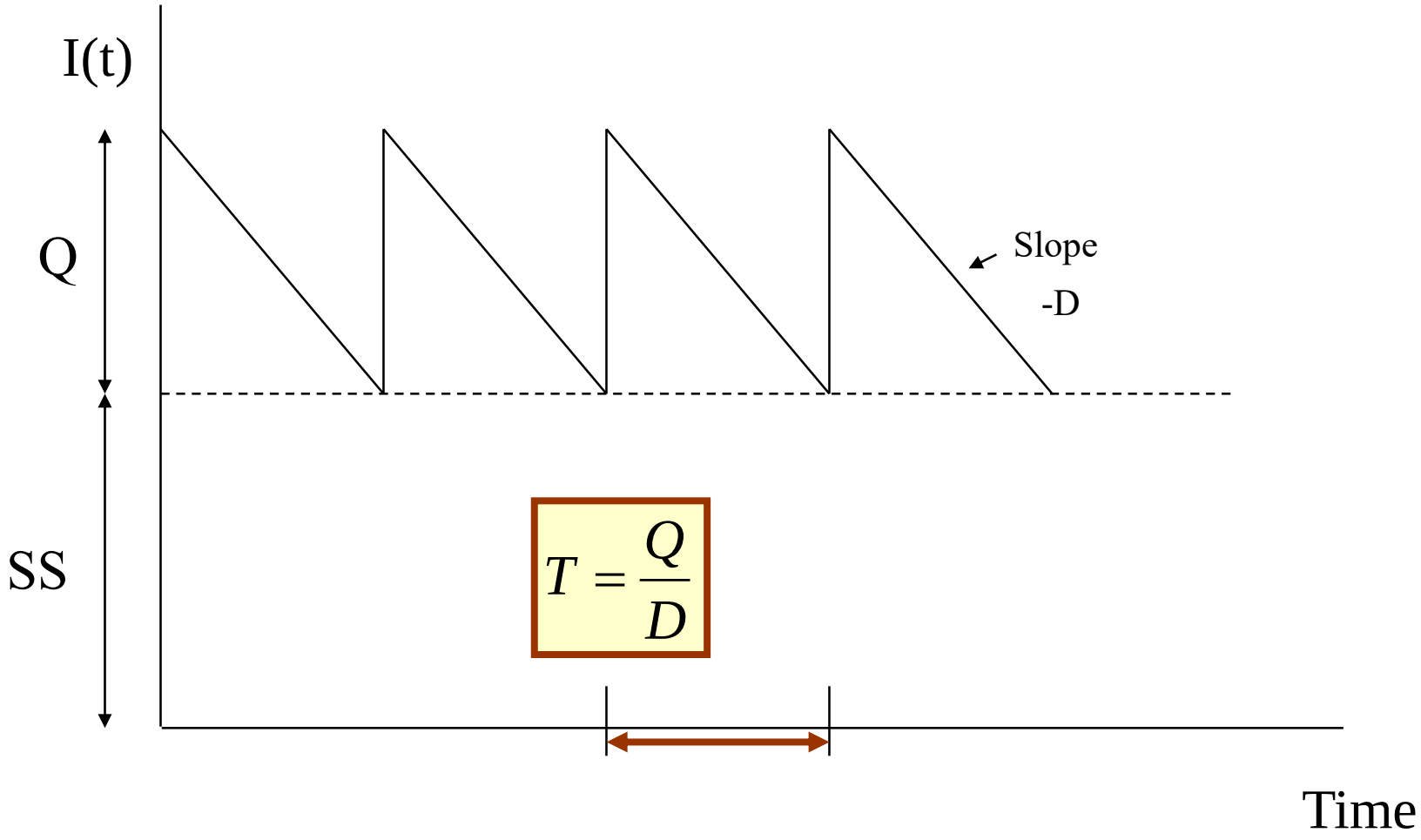
- Safety stock is the expected on-hand inventory at the end of a cycle (expected units backordered are often neglected if desired service level is high):

$$SS = R - \mu_L + E[back]$$

- Expected shortage per cycle:

$$E[back] = E[(D_L - R)^+] = \int_R^{\infty} (D_L - R) f(D_L) dD_L = n(R)$$

Expected cost function: Inventory assumption



(R, Q) model: Expected cost function

- Includes expected setup, holding, penalty, and purchasing costs per unit of time (e.g., per year).
  - Expected setup cost:

$$\frac{SD}{Q}$$

  - Expected inventory holding cost:

$$h\left(SS + \frac{Q}{2}\right) = h\left(R - \mu_L + \frac{Q}{2}\right)$$

  - Expected penalty cost:

$$b \cdot n(R) \cdot (\#\ \text{of cycles}) = b \cdot n(R) \cdot \frac{D}{Q}$$

  - Expected purchasing cost: $c \cdot D$

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(R, Q) model: Cost minimization

- Expected cost function:

$$C(R,Q) = S\frac{D}{Q} + h\left(R - \mu_L + \frac{Q}{2}\right) + b\frac{D \cdot n(R)}{Q} + cD$$

- Partial derivatives:

$$\frac{\partial C(R,Q)}{\partial Q} = -\frac{SD}{Q^2} + h\frac{1}{2} - b\frac{D \cdot n(R)}{Q^2} = 0 \Rightarrow Q = \sqrt{\frac{2D[S + b \cdot n(R)]}{h}} \quad \textbf{(1)}$$

$$\frac{\partial C(R,Q)}{\partial R} = h + b\frac{D}{Q}(-1)\big(1 - F(R)\big) = 0 \Rightarrow F(R) = \frac{bD - hQ}{bD} \quad \textbf{(2)}$$

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(R, Q) model: Solution procedure

- The optimal solution procedure requires iterating between the two equations for Q and R until convergence occurs:

1. Compute Q = EOQ
2. Substitute Q into equation (2) and compute R

$$F(R) = \frac{bD - hQ}{bD}$$

3. Use R to compute n(R)

$$n(R) = \int_R^{\infty} (D_L - R) f(D_L) dD_L$$

4. Solve for Q in equation (1)

$$Q = \sqrt{\frac{2D[S + b \cdot n(R)]}{h}}$$

5. Go back to Step 2, continue until convergence

Example: Normally distributed demand during lead time

- Consider the following case:
  - Annual demand: Normal, D = 12,000 & σ = 1,000
  - Lead time L = 1 month
  - Demand during lead time D_L : Normal, μ_L = 1,000 & σ_L = 288.67
  - Ordering cost S = $70
  - Holding cost h = $0.5 per unit per year
  - Backorder cost b = $10 per unit short

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Example: Normally distributed demand during lead time

EOQ

	Iteration			
	1	2	3	4
Q	1833,0	1925,8	1930,8	1931,1
R	1700,3	1695,1	1694,8	1694,8
z	2,4258	2,4078	2,4069	2,4068
n(R)	0,7264	0,7670	0,7692	0,7694
Q	1925,8	1930,8	1931,1	1931,1

$$F(R) = \frac{bD - hQ}{bD} \Rightarrow R = \text{norm. inv}\left(\frac{bD - hQ}{bD}; \mu_L; \sigma_L; \text{true}\right)$$

$$z = \frac{R - \mu_L}{\sigma_L} \rightarrow n(R) = \sigma_L L(z)$$

$$Q = \sqrt{\frac{2D[S + b \cdot n(R)]}{h}}$$

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Insights from the (R, Q) model

- Increasing the reorder point, R, increases safety stock but it also provides a greater buffer against stock-outs.
- Increasing the order quantity increases cycle stock but reduces ordering (or setup) costs.
- Increasing the order quantity leads to a decrease in the reorder point. (?)
- Increasing the variability of the demand tends to increase the optimal reorder point.
- Increasing the holding cost tends to decrease both the optimal order quantity and the reorder point.

The (R, Q) Model: Heuristic Approach

(R, Q) model: Heuristic Approach

- In the heuristic approach (implemented in most decision-support systems for inventory management), the parameters of the (R, Q) model are determined as follows:
  - Set order quantity (Q) equal to EOQ
  - Set reorder point (R) such that $\Pr(D_L \leq R) = CSL$

- ✓ If the backorder and holding costs are known, then the CSL can be set "optimally" (critical ratio).

Example for normally distributed demand

- Assume that demand per time unit (day, week, etc.) is independent and normally distributed with a mean = D and st.dev. = σ

- Demand during lead time is also normal with
  - Mean: $\mu_L = L \times D$
  - Standard Deviation: $\sigma_L = \sqrt{L} \times \sigma$

- In-stock probability for given R:
  - Evaluate the z-statistic for R: $z = \frac{R - \mu_L}{\sigma_L}$
  - Look up Φ(z) in the Standard Normal Distribution Function Table

(R, Q) model: Expected backorders per cycle

- Expected shortage per cycle:

$$E[back] = E[(D_L - R)^+] = \int_R^\infty (D_L - R) f(D_L) dD_L = n(R)$$

- When demand during lead time is normally distributed and for a given reorder point R:
  - Evaluate the z-statistic: $z = \frac{R - \mu_L}{\sigma_L}$
  - Look up L(z) in the Standard Normal Loss Function Table
  - $n(R) = E[back] = \sigma_L L(z)$
- Interpret n(R) as the expected number of stock-out units per cycle
- Safety stock for given R: $SS = R - \mu_L + E[short]$

(R, Q) Model: Example

- Daily demand for a certain product is normally distributed with a mean of 60 and a standard deviation of 7.
- The source of supply is reliable and maintains a constant lead time of six days.
- The cost of placing the order is $10 and annual holding costs are $0.50 per unit.
- There are no stockout costs, and unfilled orders are filled as soon as the order arrives.
- Assume sales occur over the entire 365 days of the year.
- Find the order quantity and reorder point to satisfy a 95% probability of not stocking out during the lead time.

From S. Nahmias, "Production and Operations Analysis" Ed.09 (2009)

(R, Q) Model: Computing R given CSL

- The optimal order quantity (Q) is:

$$Q = \sqrt{\frac{2DS}{H}} = \sqrt{\frac{2 \cdot 60 \cdot 365 \cdot 10}{0.50}} = 936 \text{ units}$$

- The reorder point (R) is given by:

$$R = \mu_L + z(0.95) \cdot \sigma_L = 60 \cdot 6 + 1.64 \cdot \sqrt{6} \cdot 7 = 388 \text{ units}$$

- To summarize, an order for 936 units is placed whenever the number of units remaining in inventory drops to 388

(R, Q) Model: Measuring product availability

- **Product availability**
  - A firm's ability to fill a customer's order out of available inventory
- **Type I: Cycle Service Level (CSL)**
  - Fraction of replenishment cycles that end with all customer demand being met
  - Probability of no-stock-out in a cycle
- **Type II: Fill Rate (FR)**
  - Fraction of demand satisfied from product in inventory

Example: Measuring product availability ex-post

Order Cycle	1	2	3	4	5	6	7	8	9	10	Total
Demand	180	75	235	140	180	200	150	90	160	40	1450
Stockouts	0	0	45	0	0	10	0	0	0	0	55

Type I Service Level: CSL = 0.8

Type II Service Level: FR = (1450-55)/1450 = 0.96

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(R, Q) Model: Fill Rate

- Proportion of customer demand satisfied from stock.
- E[short] is the expected shortage per cycle (expected demand in excess of reorder point in each replenishment cycle).
  - For normally distributed demand:

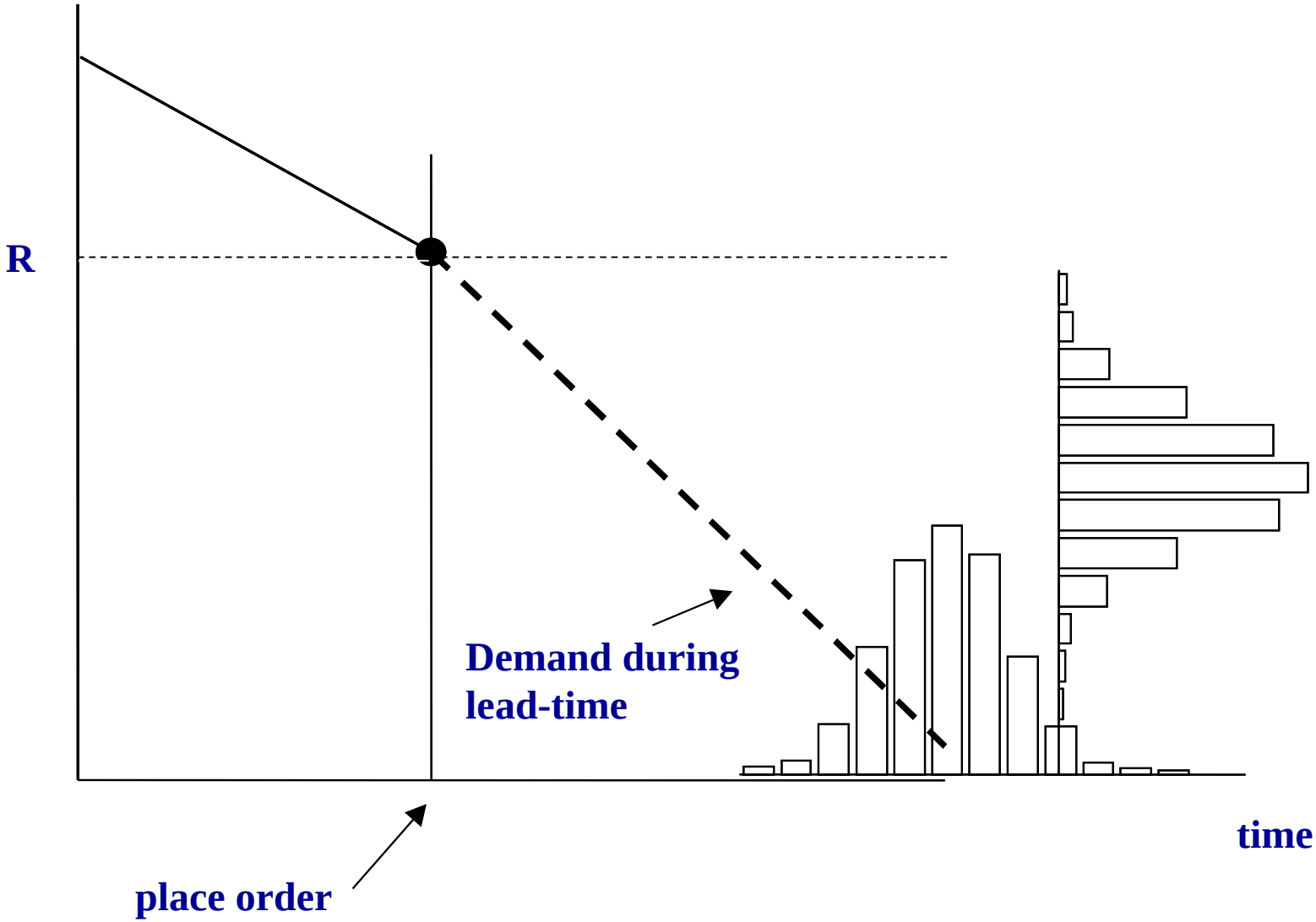
$$\mathrm{E}[\mathrm{short}] = \sigma_{\mathrm{L}} \cdot \mathrm{L}(\mathrm{z}) = \sigma_{\mathrm{L}} \cdot \left[\varphi(\mathrm{z}) - \mathrm{z}\left(1 - \Phi(\mathrm{z})\right)\right]$$

- The (expected) fill rate (FR) is then given by:

$$\mathrm{FR} = 1 - \frac{\mathrm{E}[\mathrm{short}]}{\mathrm{Q}}$$

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When both demand & lead time are uncertain



Impact of supply uncertainty

When both demand and lead time are **normally distributed** with:

D: Average demand per unit time

σ: Standard deviation of demand per unit time

L: Average lead time

s_L: Standard deviation of lead time

$$\mu_L = D \cdot L$$

$$\sigma_L = \sqrt{L \cdot \sigma^2 + D^2 \cdot s_L^2}$$

Example: Impact of supply uncertainty

Weekly demand ~ Normal (2,500, 500);

Lead time (days) ~ Normal (7, 7)

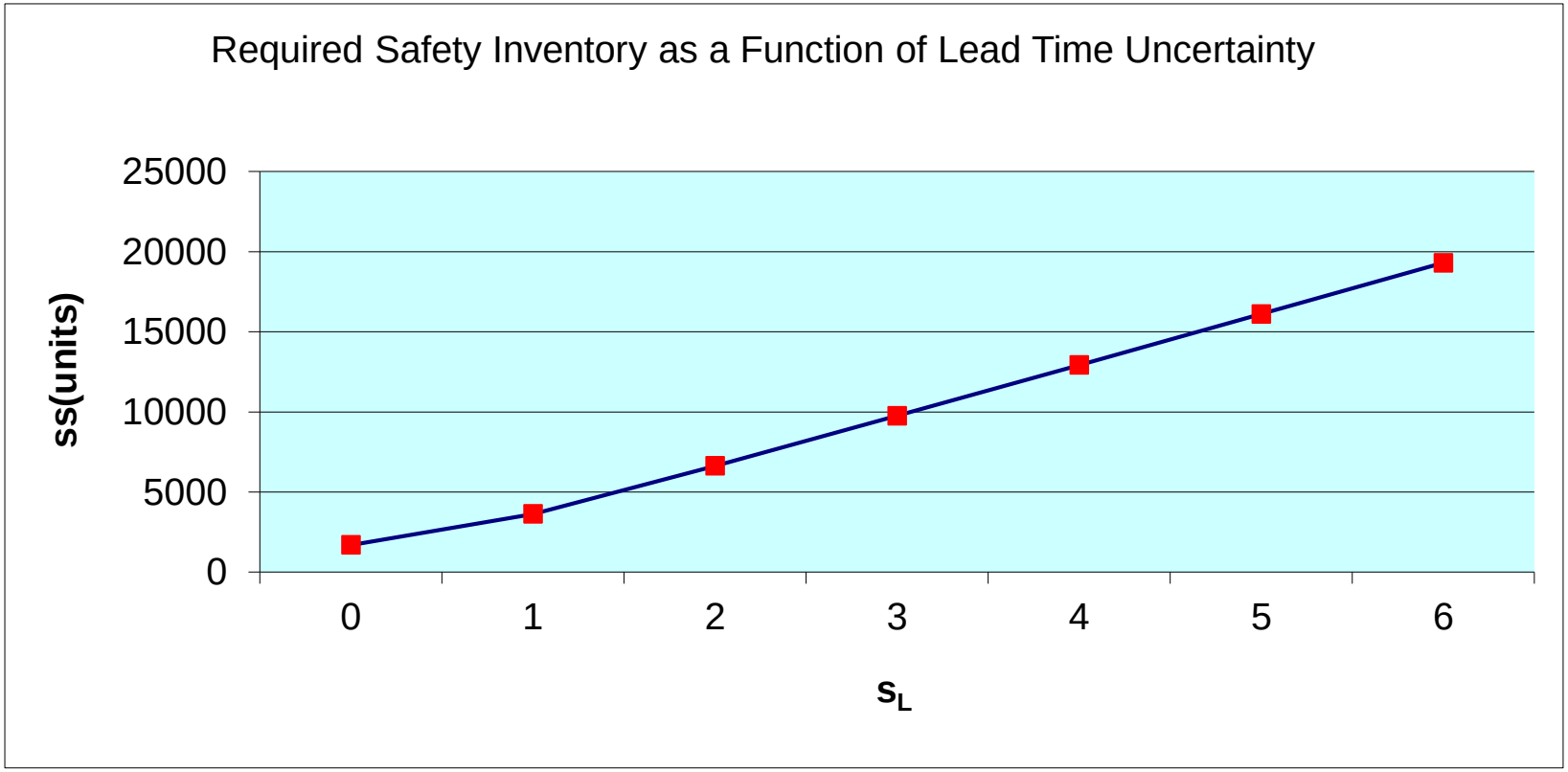
CSL = 0.90, SS =?

D_L = (D/7) x L = 2500

$\sigma_L = \sqrt{L\left(\sigma/\sqrt{7}\right)^2 + \left(D/7\right)^2 s_L^2} = \sqrt{7 \cdot 189^2 + 357^2 \cdot 7^2} = 2{,}548.5$

SS = $\Phi^{-1}(CSL)\sigma_L$ = NORM.S.INV(0.90)*2,548.5 = 3,266

Impact of supply uncertainty



A reduction in supply uncertainty can help dramatically reduce safety inventory without hurting product availability.

Impact of desired product availability & uncertainty on safety inventory

- If desired product availability (CSL or FR) increases, then the required safety inventory increases.
- If demand uncertainty (σ_D) increases, then the required safety inventory increases.
- Managerial levers to reduce safety inventory without reducing product availability:
  - Reduce supplier lead time, L (better relationships with suppliers).
  - Eliminate lead time volatility.
  - Reduce uncertainty in demand, σ_D (better forecasts, better information collection and use).

(R, Q) model: Summary

- Inventory position constantly monitored (inventory position –and not inventory level– is used for placing an order).
- Stockout occurs if demand during lead time exceeds the reorder point.
- Simple system; suppliers like the placement of constant order quantities; however, the timing of order placement is now stochastic.
- A heuristic solution that is often used in practice sets Q = EOQ and R such that $\Pr(D_L \leq R) = CSL$
  - If the backorder and holding costs are known, then the CSL can be set optimally (critical ratio).

Periodic vs. continuous review systems

- In periodic review the safety stock is larger as it covers the uncertainty over T+L time units; i.e., T units of time more than in the continuous case.
- Primary advantages of periodic systems:
  - Convenience
  - Orders can be combined
- Primary advantages of continuous systems:
  - Fixed lot sizes can facilitate quantity discounts
  - Keeps constant count of inventory
- Disadvantages of continuous systems:
  - Higher record keeping cost
  - Time of delivery is random

Readings

- Chapter 11 (sections 11.1, 11.2, 11.3 and 11.5) and Chapter 12 (sections 12.1 and 12.2) from Chopra & Meindl, 4th ed.

Thank you!

APPENDIX

The (R, Q) Model:

Approximate Formulation of the Lost Sales Case

(R, Q) model: Expected lost sales per cycle

- Demand during lead time is a random variable, D_L, with
  - PDF(density function) f(x) and CDF (distribution function) F(x)
  - Mean = μ_L and Standard Deviation = σ_L
- Expected lost sales per cycle:

$$E[\text{lost}] = E[(D_L - R)^+] = \int_R^\infty (D_L - R) f(D_L) dD_L = n(R)$$

- Safety stock is the expected on-hand inventory at the end of a cycle (now the expected number of lost sales is also considered)

$$SS = R - \mu_L + n(R)$$

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(R, Q) model with lost sales: Expected cost function

- Includes expected setup, holding, lost sales penalty, and purchasing costs per unit of time (e.g., per year), assuming that the average length of time that the system is out of stock is small enough to be neglected:
  - Expected setup cost: $\frac{SD}{Q}$
  - Expected inventory holding cost:

$$h\left(SS + n(R) + \frac{Q}{2}\right) = h\left(R - \mu_L + n(R) + \frac{Q}{2}\right)$$

  - Expected penalty cost:

$$b \cdot n(R) \cdot (\#\ \text{of cycles}) = b \cdot n(R) \cdot \frac{D}{Q}$$

  - Expected purchasing cost: $c \cdot D$

(R, Q) model with lost sales: Cost minimization

- Expected cost function:

$$C(R,Q) = S\frac{D}{Q} + h\left(R - \mu_L + \frac{Q}{2}\right) + \left(h + b\frac{D}{Q}\right)n(R) + cD$$

- Partial derivatives:

$$\frac{\partial C(R,Q)}{\partial Q} = -\frac{SD}{Q^2} + h\frac{1}{2} - b\frac{D \cdot n(R)}{Q^2} = 0 \Rightarrow Q = \sqrt{\frac{2D[S + b \cdot n(R)]}{h}} \quad \textbf{(1)}$$

$$\frac{\partial C(R,Q)}{\partial R} = h + \left(h + b\frac{D}{Q}\right)(-1)\big(1 - F(R)\big) = 0 \Rightarrow F(R) = \frac{bD}{bD + hQ} \quad \textbf{(2)}$$

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(R, Q) model with lost sales: Solution procedure

- The optimal solution procedure requires iterating between the two equations for Q and R until convergence occurs:

1. Compute Q = EOQ
2. Substitute Q into equation (2) and compute R

$$F(R) = \frac{bD}{bD+hQ}$$

3. Use R to compute n(R)

$$n(R) = \int_R^{\infty} (D_L - R) f(D_L) dD_L$$

4. Solve for Q in equation (1)

$$Q = \sqrt{\frac{2D[S+b \cdot n(R)]}{h}}$$

5. Go back to Step 2, continue until convergence